

Regression Algorithm – Insurance Charges Prediction

❖ Problem Statement:

The client wants to predict the insurance charges of a person based on several parameters such as age, sex, BMI, number of children, and smoking status. The provided dataset contains **1338 rows and 6 columns**.

❖ Pre-processing:

The dataset contains categorical/string values such as **sex** and **smoker**. I used **One-Hot Encoding** to convert the categorical values into numerical values so that they can be used by machine learning algorithms.

❖ Algorithms Used:

- Random Forest Regressor
- Decision Tree Regressor
- Multiple Linear Regression

❖ Evaluation Metric:

I used the **R² (R-squared) score** to compare the performance of the regression models.

➤ Support Vector Machine:

Without standardization:

```
: from sklearn.metrics import r2_score  
r_score=r2_score(y_test,y_pred)  
r_score
```

```
: -0.12207668380229886
```

With Standardization:

```
[25]: from sklearn.metrics import r2_score
      r_score=r2_score(y_test,y_pred)
      r_score
```

```
[25]: -0.080959968427891
```

➤ Random Forest:

```
: from sklearn.model_selection import train_test_split
  X_train,X_test,y_train,y_test=train_test_split(independent,dependent,test_size=0.3,random_state=0)

: from sklearn.ensemble import RandomForestRegressor
  regressor=RandomForestRegressor(n_estimators=100,random_state=0)
  regressor=regressor.fit(X_train,y_train)

C:\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed as a
e shape of y to (n_samples,), for example using ravel().
  return fit_method(estimator, *args, **kwargs)

: y_pred=regressor.predict(X_test)

  from sklearn.metrics import r2_score
  r_score=r2_score(y_test,y_pred)
  r_score

: 0.8539781318598829
```

➤ Decision Tree:

```

: from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(independent,dependent,test_size=0.3,random_state=0)

: from sklearn.ensemble import RandomForestRegressor
regressor=RandomForestRegressor(n_estimators=100,random_state=0)
regressor=regressor.fit(X_train,y_train)

C:\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed
e shape of y to (n_samples,), for example using ravel().
    return fit_method(estimator, *args, **kwargs)

: y_pred=regressor.predict(X_test)

from sklearn.metrics import r2_score
r_score=r2_score(y_test,y_pred)
r_score

: 0.8539781318598829

```

➤ Multi Linear Regression:

```

: from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(independent,dependent,test_size=0.30,random_state=0)

: from sklearn.linear_model import LinearRegression
regressor=LinearRegression()
regressor.fit(X_train,y_train)

: ▼ LinearRegression ⓘ ?
  ▶ Parameters
  ▶ Fitted attributes

: y_pred=regressor.predict(X_test)
from sklearn.metrics import r2_score
r_score=r2_score(y_test,y_pred)
r_score

: 0.7894790349867009

```

❖ Best Model:

The **Random Forest Regressor** performed the best, with an **R² score of 0.853**.

❖ Conclusion:

Among the three regression algorithms, the **Random Forest Regressor** gave the highest R² score of **0.853**. Therefore, Random Forest is the best-performing model for predicting insurance charges in this dataset. An R² score of 0.853 means that the model

explains approximately **85.3% of the variation in insurance charges** based on the input feature